Homework 3

Due : Sunday, May 24 at 11 : 59 pm

Honor Code

Write and sign the following statement:

> "I certify that all solutions in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted."

1 Evaluation Metrics

Assume that a logistic regression classifier has been trained to estimate the probability that an input belongs to the positive class, denoted as $\hat{p}(y = 1|x)$, for each sample in a test set. The table below presents the actual class labels together with the model's probability estimates for a small collection of test instances:

ID	**True** y	$\hat{p}$
1	1	0.94
2	1	0.83
3	0	0.71
4	0	0.62
5	0	0.59
6	1	0.48
7	0	0.39
8	0	0.36
9	1	0.33
10	0	0.17

The model assigns a predicted label $\hat{y} = 1$ whenever $\hat{p} \geq \tau$, where $\tau \in (0, 1)$ denotes the decision threshold for binary classification.

(a) Explain accuracy, precision, and recall for binary classification and their differences.

(b) For each decision threshold $\tau \in \{0.3, 0.5, 0.8\}$, calculate True Positive (TP), False Positive (FP), True Negative (TN), False Negative (FN). Then, calculate accuracy, precision, and recall.

(c) How would you determine the most appropriate decision threshold for this dataset, given the following scenarios?

- In a medical screening setting, where $y = 1$ means a patient truly has the disease and $\hat{y} = 1$ means the disease is detected and treatment is administered, which threshold from part (b) would you select? Briefly explain your reasoning.
- How would your choice differ in a spam-filtering scenario, where $y = 1$ denotes a spam email and $\hat{y} = 1$ means the email is classified as spam and automatically deleted?

2 Sigmoid vs Softmax

In multi-class classification problems with K classes, the Softmax function is commonly used to convert raw logits into a probability distribution. For binary classification ($K = 2$), the Sigmoid (or Logistic) function is the standard choice.
Let the raw logits for a binary classification problem be a vector $\mathbf{z} = [z_0, z_1]$, where z_0 corresponds to class 0 and z_1 corresponds to class 1.

(a) Write down the expression for the probability of class 1, $P(y = 1|\mathbf{z})$, using the Softmax function.

(b) Show algebraically that this expression is equivalent to applying the Sigmoid function, $\sigma(x) = \frac{1}{1+e^{-x}}$, to a scalar input derived from the logits z_0 and z_1.

(c) Based on this equivalence, explain why binary classification models typically use a single output neuron with a Sigmoid activation rather than two output neurons with a Softmax activation.

3 Decision Boundary

Suppose a logistic regression classifier has been trained, producing class probabilities given by

$$z(x) = \sigma(w^T x + b).$$

The final class prediction is made using the decision rule

$$y(x) = \mathbb{1}[z(x) > 0.5].$$

Demonstrate that this decision rule corresponds to a linear decision boundary in the input space.

4 Rewriting Objective Function

In the lecture, we represented the target labels for logistic regression as $t^{(i)} \in \{0, 1\}$. In this exercise, you will derive an equivalent formulation when the labels are instead encoded as $\tilde{t}^{(i)} \in \{-1, 1\}$.

Given a training set $D_N = \{(x^{(i)}, t^{(i)}\}$ with $t^{(i)} \in \{0, 1\}$, logistic regression is defined by the following relationships:

$$z = w^T x + b, \qquad y = \sigma(z)$$

with the loss function

$$L(y, t) = -t \log(y) - (1 - t) \log(1 - y).$$

(a) Express the corresponding optimization problem over the training data by removing the intermediate variables y and z. Your final objective function should be written solely in terms of the parameters w and b, along with the dataset D.

(b) Demonstrate that when the target labels are encoded as $\tilde{t}^{(i)} \in \{-1, 1\}$, the training objective can be written as the following optimization problem:

$$\min_{w,b} \sum_{i=1}^{N} \log\left(1 + \exp\{-\tilde{t}^{(i)}(w^T x^{(i)} + b)\}\right)$$

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BG,

1)

a)

True Positive (TP): The model correctly predicts the positive class.

True Negative (TN): The model correctly predicts the negative class.

False Positives (FP): The model incorrectly predicts the positive class.

False Negatives (FN): The model incorrectly predicts the negative class.

Accuracy: Measures the overall correctness of the model. It is the ratio of all correct predictions to the total number of cases.

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}}$$

Precision: Measures the quality of positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall; Measures the ability to detect actual positives.

$$Recall = \frac{TP}{TP+FN}$$

Key Difference

- Accuracy is best for balanced datasets. It is highly misleading for imbalanced data, where a model might just predict the majority class.
- Precision is prioritized when False Positives are costly.
- Recall is prioritized when False Negatives are dangerous.

b)

For $\gamma = 0.3$

$TP = 4,\ FP = 5,\ TN = 1,\ FN = 0$

$$Accuracy = \frac{5}{10} = 0.50$$

$$Precision = \frac{4}{9} \approx 0.444$$

$$Recall = \frac{4}{4} = 1.00$$

For $\gamma = 0.5$

$TP = 2,\ FP = 3,\ TN = 3,\ FN = 2$

$$Accuracy = \frac{5}{10} = 0.5$$

$$Precision = \frac{2}{5} = 0.4$$

Recall $= \frac{2}{4} = 0.5$

For $\tau = 0.8$

TP = 2, FP = 0, TN = 6, FN = 2

Accuracy $= \frac{8}{10} = 0.8$

Precision $= \frac{2}{2} = 1.0$

Recall $= \frac{2}{4} = 0.50$

c)

Medical Screening Scenario:

Selected Threshold $\tau = 0.3$

In a medical context, the cost of a False Negative is severe. Therefore, we must prioritize maximizing Recall to catch as many true cases as possible.

Spam Filtering Scenario:

Selected Threshold $\tau = 0.8$

In this scenario, classifying a legitimate email as spam results in its automatic deletion. Because the cost of a False Positive is unacceptably high, we must prioritize maximizing Precision.

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2)

a)

Given the raw logits vector $z = \{2_0, 2_1\}$ for a binary classification problem, the general

Softmax function calculates the probability for class i as:

$$P(y=i|z) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

For class 1 ($y=1$), the expression using the Softmax function is:

$$P(y=1|z) = \frac{e^{z_1}}{e^{z_0}+e^{z_1}}$$

b)

$$P(y=1|z) = \frac{e^{z_1}}{e^{z_0}+e^{z_1}} = \frac{\frac{e^{z_1}}{e^{z_1}}}{\frac{e^{z_0}}{e^{z_1}}+\frac{e^{z_1}}{e^{z_1}}}$$

$$= \frac{1}{e^{z_0-z_1}+1} = \frac{1}{1+e^{-(z_1-z_0)}}$$

$$\sigma(x) = \frac{1}{1+e^{-x}} \qquad x = z_1-z_0$$

c)

Since both approaches are mathematically equivalent for binary classification, a single Sigmoid output is preferred for efficiency

Parameter Efficiency: A single output neuron halves the number of weights required in the final layer. This reduces memory usage and computational cost.

No Redundancy: In binary classification, the probabilities must sum to 1. Calculating both probabilities seperately via Softmax is redundant. Sigmoid computes

only the necessary positive class probability,

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3)

$z(x) = 0.5$

$\sigma(w^T x + b) = 0.5$

$$\frac{1}{1+e^{-(w^T x+b)}} = 0.5$$

$$1 + e^{-(w^T x+b)} = 2$$

$$e^{-(w^T x+b)} = 1 \Rightarrow -(w^T x+b) = \ln(1) \Rightarrow -(w^T x+b) = 0$$

$$w^T x + b = 0$$

The resulting equation, $w^T x + b = 0$, is the standart mathematical definition of a linear hyperplane. This demonstrates that the decision rule corresponds to a strictly linear decision boundary in the input space x.

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4)

a)

$$L(y,t) = -t\log(y) - (1-t)\log(1-y)$$

$$y = \sigma(2) = \frac{1}{1+e^{-2}}$$

$$\log(y) = \log\left(\frac{1}{1+e^{-2}}\right) = -\log(1+e^{-2})$$

$$1-y = 1 - \frac{1}{1+e^{-2}} = \frac{e^{-2}}{1+e^{-2}} = \frac{1}{1+e^{2}}$$

$$\log(1-y) = \log\left(\frac{1}{1+e^{2}}\right) = -\log(1+e^{2})$$

$$L(z,t) = -t(-\log(1+e^{-z})) - (1-t)(-\log(1+e^{z}))$$

$$L(z,t) = t\log(1+e^{-z}) + (1-t)\log(1+e^{z})$$

Substitute $z^{(i)} = w^T x^{(i)} + b$ for each data point i, and sum over all N examples in the training set D_N:

$$J(w,b) = \sum_{i=1}^{N}\left[t^{(i)}\log(1+e^{-(w^T x^{(i)}+b)}) + (1-t^{(i)})\log(1+e^{w^T x^{(i)}+b})\right]$$

b)

Let $z^{(i)} = w^T x^{(i)} + b$. From part a, the loss function for a single example with true label $t \in \{0,1\}$ is:

$$L(z^{(i)}, t^{(i)}) = t^{(i)}\log(1+e^{-z^{(i)}}) + (1-t^{(i)})\log(1+e^{z^{(i)}})$$

Let's evaluate this expression for the two possible classes and map them to the new encoding $\tilde{t} \in \{-1,1\}$

Positive Class ($t^{(i)} = 1 \Rightarrow \tilde{t}^{(i)} = 1$)

$$L = 1 \cdot \log(1+e^{-z^{(i)}}) + 0 = \log(1+e^{-z^{(i)}})$$

Notice that we can rewrite this using $\tilde{t}^{(i)} = 1$

$$L = \log(1+\exp\{-\tilde{t}^{(i)} z^{(i)}\})$$

Negative Case ($t^{(i)} = 0 \Rightarrow \tilde{t}^{(i)} = -1$)

$$L = 0 + 1 \cdot \log(1+e^{z^{(i)}}) = \log(1+e^{z^{(i)}})$$

$$\tilde{t}^{(i)} = -1$$

$$L = \log(1+\exp\{-\tilde{t}^{(i)} z^{(i)}\})$$

In both cases, the piecewise loss function simplifies perfectly to a single, unified expression:

$\log(1 + \exp\{-\tilde{t}^{(i)} z^{(i)}\})$

To train the logistic regression model, we want to find the parameters w and b that minimize the total loss over the entire dataset of N examples. Summing this unified loss function over $i=1$ to N yields the exact optimization problem provided:

$$\min_{w,b} \sum_{i=1}^{N} \log(1 + \exp(-\tilde{t}^{(i)}(w^T x^{(i)} + b))\}$$